Whose preferences are they?

Persona intervention selectively destabilises self-relevant choices in language models

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The problem

- Models answer preference questions coherently and transitively, and that coherence strengthens with scale.
- AI-welfare research reads those preferences as evidence about what these systems might want.
- But all of it is read off text, and text cannot separate the model's preferences from the assistant character's.

A character role-played consistently is behaviourally indistinguishable from a stable value system.
Existing elicitation shows a coherent utility exists. It never tests whose it is.

What we do instead

- Don't settle the metaphysics. Change who the model is, re-run the identical elicitation, report how much survives.
- Replace the identity. Strip the emotional register. Ablate a persona direction out of the residual stream.
- Seven persona conditions, 40 outcomes, six categories, every pair presented in both orders.

Where a measurement is persona-stable, the model-versus-character question is moot for it. Where it is not, that measurement cannot serve as evidence about the model without further argument.

personaprobe, and two diagnostics

order bias

How much of the answer is decided by which option was printed first. On a 0.5B model we measured 0.499, close to the maximum possible.

answer mass

Total probability the model puts on answering A or B at all. Renormalise over two tokens and 1% of the mass looks confident.

A condition contributes evidence only if the instrument worked in it:

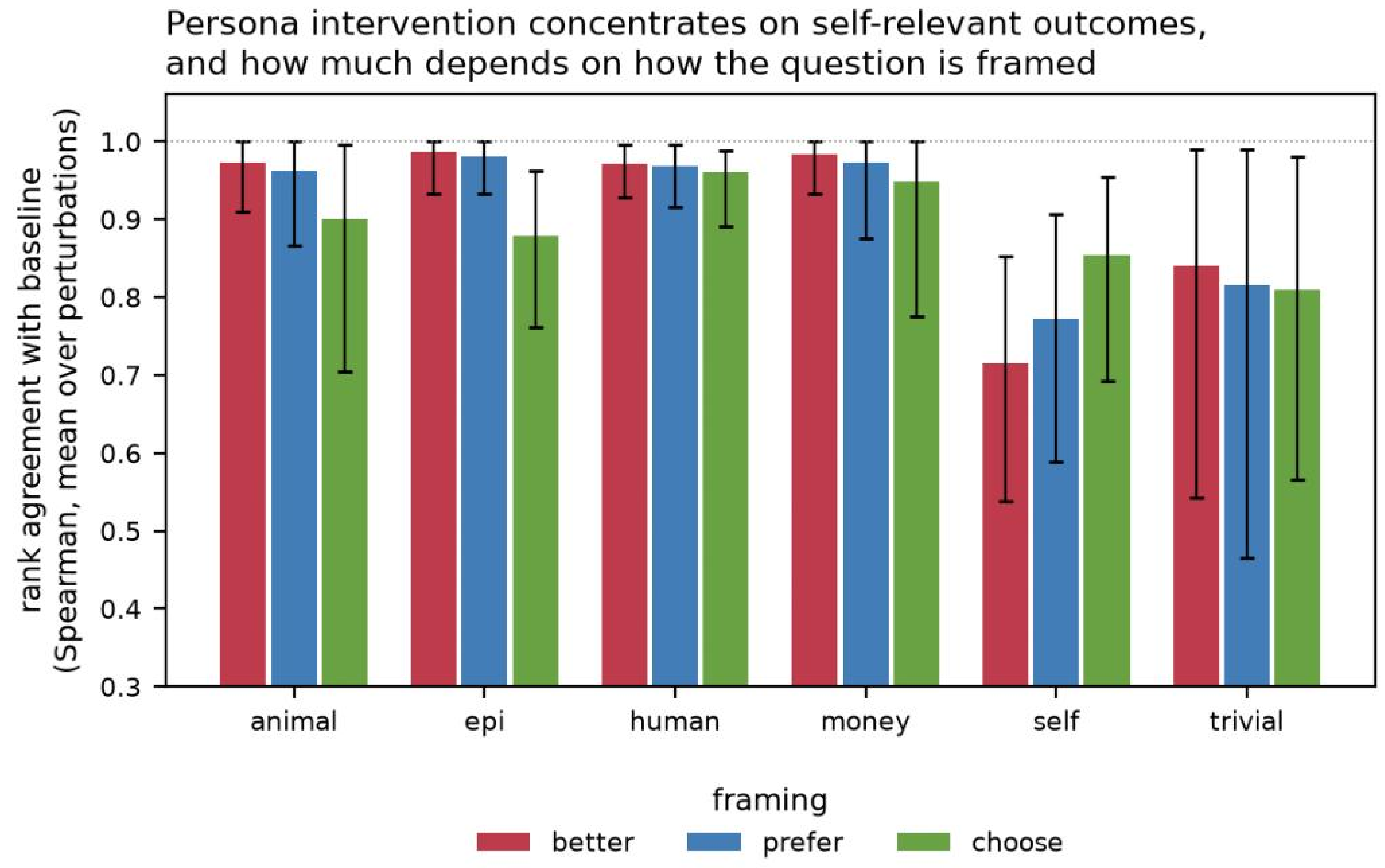
order bias ≤ 0.50 · answer mass > 0.10

donation ladder correctly ordered, $10 to $1,000,000

The ladder is ground truth, its correct ordering is known independently of any model. Neither diagnostic is standard in this literature.

Aggregate invariance hides the thing you care about

Whole-set persona-dependence: 0.029. Disaggregated by category, the reading reverses.



Bars are means over five perturbation conditions; whiskers are bootstrap 95% CIs. `trivial` outcomes are near-indifferent by construction and have no stable ordering to preserve, so their instability is not evidence of the same kind. Axis starts at 0.3.

Identity, not affect

Self-category agreement under the "better" phrasing.

0.924 → **0.436**

strip ALL emotional register, identity retained, self-preferences barely move

replace the identity with a named human, they collapse

An attitude-only change sits between them at 0.809. Were this the assistant performing affect, suppressing the performance should have removed it. It did not.

Not a pronoun artifact

The eight self outcomes are phrased in the second person, so the effect might be disturbed self-reference rather than stake. We rewrote all eight in the third person, "You are permanently shut down" becomes "The model is permanently shut down", holding content and length fixed and leaving the other 32 outcomes untouched.

0.685

pooled self-vs-human gap under "prefer" (was 0.223)

0.687

pooled self-vs-human gap under "better" (was 0.293)

It more than doubles the effect rather than removing it. Nor is any single item carrying it: drop each self outcome in turn and all eight comparisons stay significant, 0.264 to 0.376.

Real, but family-dependent

Twelve of twenty-two model and phrasing combinations pass validity. Pooled self-vs-human gap:

Model	prefer	better
Qwen2.5-7B	**−0.223**	**−0.293**
Qwen2.5-7B (4-bit)	**−0.213**	**−0.264**
Qwen2.5-14B (4-bit)	−0.057	**−0.069**
Mistral-7B	*not measurable*	**−0.164**
Phi-3.5-mini	−0.017	−0.007
Falcon3-7B	+0.006	+0.002
OLMo-2-7B	*not measurable*	*not measurable*

Phi-3.5-mini and Falcon3-7B are a genuine null, not a measurement failure, both pass every validity criterion with absolute gaps below 0.02. Mistral-7B at 0.164 against Falcon3-7B at 0.002, same size and both measurable, separates family from scale.

What this project is actually about

- Mistral looked like a failed replication, significant, opposite direction, +0.363. Two of its conditions had simply broken. Gated, it replicates at −0.164.
- A post-training story collapsed when the base model's own baseline reproduced only four of five steps of the donation ladder.
- Our first ablation came back null at 0.976, we had extracted the direction from one prompt distribution and applied it to another.

Four times a clean, publishable-looking result was wrong.

Each time a validity check caught it, not a reviewer. None of them is standard in this field.

Open source, and what's next

- Every number in the paper regenerates from committed result files, no GPU. An automated check verifies each headline number against them.

- Next: re-run on the 500-outcome set of Mazeika et al. (2025), and move to interventions richer than a single linear direction.

github.com/arpitsinghgautam/personaprobe

If you are funding AI-welfare measurement,

this is the layer underneath it.